Five emission calculations

One whole episode, one pollutant. O, S, R, M and A identify the existing calculations.

Notation and units. Bin i has speed v_i and count $d_i = q_i \Delta t$, with $\Delta t = 1/12$ h. $D = \sum_i d_i$; L is link length; C is hourly capacity; $T_f = L/v_f$. H denotes vehicle-hours, P episode duration and T per-vehicle travel time. $r_0(v)$ is the zero-acceleration lookup rate in g/(veh h); E is in grams. Counts and capacity use the same lane basis.

Shared fitted quantities for R, M and A.

$$x = \frac{D}{C}, \qquad \widehat{P} = \max(f_d x^n, x), \qquad \widehat{w}_{\max} = T_f f_p \widehat{P}^s, \qquad \widehat{\bar{w}} = \theta \widehat{w}_{\max}, \qquad H_{\text{pred}} = D(T_f + \widehat{\bar{w}}).$$

f_d, n, f_p, s, θ are calibrated by directional corridor and period, with the existing sparse-period fallback.

O Recorded-speed reference

Assign each bin's counted vehicles the complete-link travel time L/v_i, then sum:

$$H_i = d_i \frac{L}{v_i}, \qquad H_{\text{obs}} = \sum_i H_i, \qquad \boxed{E_O = \sum_i H_i r_0(v_i).}$$

The five-minute interval supplies the count; emission exposure is L/v_i, not five minutes.

S Episode mean-speed estimate

Use the count-weighted harmonic mean, preserving the reference's total vehicle-hours:

$$\bar{v}_{\text{obs}} = \frac{D}{\sum_i d_i/v_i} = \frac{DL}{H_{\text{obs}}}, \qquad \boxed{E_S = H_{\text{obs}}\, r_0(\bar{v}_{\text{obs}}).}$$

R Reconstructed whole-link speeds

For normalized entry time $u \in [0,1]$, define $w(u) = \widehat{w}_{\max} h(u)$ and $v_R(u) = L/[T_f + w(u)]$. The symmetric profile is

$$h(u) = \begin{cases} [4u(1-u)]^a, & \theta \le 2/3, \quad \int_0^1 h(u)\,du = \theta \text{ determines } a, \\ 1 - |2u-1|^b, & \theta > 2/3, \quad b = \theta/(1-\theta). \end{cases}$$

The entering-vehicle weight is $dN = D[1 + (\widehat{w}_{\max}/\widehat{P})h'(u)]\,du$. Thus

$$\boxed{E_R = \int_0^1 [T_f + w(u)]\, r_0\!\left(\frac{L}{T_f + w(u)}\right) dN.}$$

The integral uses 384 quadrature points; $\int dN = D$ and $\int [T_f + w(u)] dN = H_{\text{pred}}$. These are trip-average speeds by entry time, not individual operating-speed trajectories.

M Two-speed lookup

Infer the queue speed from discharge and the triangular fundamental diagram:

$$\widehat{\mu} = \frac{D}{\widehat{\bar{P}}}, \qquad \omega = \frac{C}{k_j - C/v_f}, \qquad \widehat{v}_q = \frac{\widehat{\mu}}{k_j - \widehat{\mu}/\omega}, \qquad k_j = 220 \text{ veh/mi/lane}.$$

Allocate mean travel time between two operating speeds while preserving distance:

$$\bar{T}_q = \frac{v_f \widehat{\bar{w}}}{v_f - \widehat{v}_q}, \qquad \bar{T}_{\text{free}} = T_f - \frac{\widehat{v}_q \widehat{\bar{w}}}{v_f - \widehat{v}_q}, \qquad \boxed{E_M = D[\bar{T}_{\text{free}} r_0(v_f) + \bar{T}_q r_0(\widehat{v}_q)].}$$

Here $\bar{T}_{\text{free}} + \bar{T}_q = T_f + \widehat{\bar{w}}$ and $v_f \bar{T}_{\text{free}} + \widehat{v}_q \bar{T}_q = L$.

A Two-speed cubic

Keep M's speeds and times; replace the lookup rate by its fitted cubic:

$$r_c(v) = c_0 + c_1 v + c_2 v^2 + c_3 v^3, \qquad \boxed{E_A = D[\bar{T}_{\text{free}} r_c(v_f) + \bar{T}_q r_c(\widehat{v}_q)].}$$

Evaluation: episode-total WAPE. For method $m \in \{S, R, M, A\}$ and the same $N = 137$ episodes,

$$\boxed{\text{WAPE}_m = 100 \frac{\sum_{k=1}^{N} |E_{m,k} - E_{O,k}|}{\sum_{k=1}^{N} E_{O,k}}.}$$

Absolute errors are calculated before pooling episodes. O is a detector-speed-based calculation, not measured emissions or observed vehicle trajectories. All five calculations omit acceleration effects.

Results on the common episode sample

137 average-weekday episodes: 20 Arizona I-10 and 117 California PeMS.

Table 1. CO_2 totals, travel-time exposure and episode-total error.

Calculation	Vehicle-hours (veh h)	Effective rate* kg/(veh h)	Total CO_2 (kg)	WAPE (%)
O: recorded speeds	4,537.48	9.063	41,123.55	Reference
S: episode mean speed	4,537.48	8.646	39,231.21	5.94
R: reconstructed speeds	4,453.44	9.623	42,856.84	10.13
M: two-speed lookup	4,453.44	11.150	49,655.91	22.38
A: two-speed cubic	4,453.44	10.967	48,838.84	20.32

*Effective rate is summed emissions divided by summed vehicle-hours. It is not a lookup rate at a pooled speed. WAPE is based on individual episode errors, not the error of the summed totals.

Table 2. Episode-total WAPE against O for all four pollutants (%).

Calculation	CO_2	NO_x	CO	HC
S: episode mean speed	5.94	12.43	6.12	4.24
R: reconstructed speeds	10.13	13.05	8.27	9.52
M: two-speed lookup	22.38	47.73	11.19	12.61
A: two-speed cubic	20.32	42.71	9.82	11.65

Reading the comparison

S preserves observed exposure. It retains O's vehicle-hours and total distance exactly, changing only how emission rates are averaged. R, M and A use fitted vehicle-hours instead. The fitted vehicle-hours have an episode WAPE of 14.77%, although their summed bias is only −1.85%.

M versus R isolates a speed-representation difference. They share the same fitted vehicle-hours and total distance, but M produces 15.9% more CO_2 and 32.1% more NO_x in total. R applies rates to whole-link trip-average speeds; M applies rates to free-flow and queue operating speeds.

Why one mean speed can be close to the reference

Let $a_i = H_i/H_{\text{obs}}$ within an episode. Then $\sum_i a_i = 1$ and $\bar{v}_{\text{obs}} = \sum_i a_i v_i$, giving

$$E_O - E_S = H_{\text{obs}} \left[\sum_i a_i r_0(v_i) - r_0\left(\sum_i a_i v_i \right) \right].$$

If $r_0(v) = a + bv$ over the occupied speed range, S equals O exactly despite speed variability. The actual stepped lookup is nonlinear, so the discrepancy depends on its rate changes and the speed distribution.

Interpretation limit. Lower WAPE means closer agreement with O. It does not establish greater accuracy against actual vehicle trajectories or measured emissions. A single-speed comparator using the same fitted exposure, $E_{S,\text{pred}} = H_{\text{pred}} r_0(DL/H_{\text{pred}})$, is not included in these reported results.

Source: archived average-weekday episode results and emission metrics (22 September 2026), filtered to the common five-method sample. Existing outputs were checked and summarized; no experiments were rerun. Formulas follow the retained implementation. Totals are passenger-car-equivalent estimates on the source per-lane basis.

Detector 78: worked link example

Arizona I-10 westbound, January 2016 average weekday.

Scope. This is the standalone workbook example from the supplied sensor-78 data: 16 complete weekdays, 57 bins from 14:45 inclusive to 19:30 exclusive. The window is fixed for explanation, not newly detected. These inputs differ from the archived paper sample; the case is not one of the 137 observations summarized on page 2. Supplied counts are assumed already per lane (multiplier 1).

Inputs and fitted traffic quantities

Input	Value	Calculated quantity	Value
Link length L	1.040 mi	Demand D	4,789.25 veh/lane
Free-flow speed v_f	64.161 mph	D/C	3.189 h
Capacity C	1,501.616 veh/h/lane	Predicted duration $\widehat{P}$	4.449 h
Jam density k_j	220 veh/mi/lane	Queue speed $\widehat{v}_q$	13.611 mph
Observed duration	4.750 h	Predicted mean delay	1.496 min
Observed vehicle-hours	211.434 veh h/lane	Predicted vehicle-hours	197.040 veh h/lane
Observed mean speed	23.557 mph	Predicted mean trip speed	25.278 mph

Calibration: $f_d = 1.693177$, $n = 0.833031$, $f_p = 0.308070$, $s = 1.635098$, $\theta = 0.434829$ (PM). The reconstructed shape exponent is $a = 3.396386$.

How the CO_2 totals are obtained

$$E_O = \sum_{i=1}^{57} d_i(L/v_i)r_0(v_i)/1000 = 1,743.79 \text{ kg/lane},$$
$$E_S = 211.434 \times 6,931.846/1000 = 1,465.63 \text{ kg/lane},$$
$$E_R = \sum_{j=1}^{384} N_j[T_f + w(u_j)]r_0(v_R(u_j))/1000 = 1,657.64 \text{ kg/lane},$$
$$E_M = 4,789.25[0.00949555 \times 20,615.635 + 0.03164658 \times 6,931.846]/1000 = 1,988.14 \text{ kg/lane},$$
$$E_A = 4,789.25[0.00949555 \times 18,172.460 + 0.03164658 \times 6,680.499]/1000 = 1,838.94 \text{ kg/lane}.$$

The numerical rates are in g/(veh h), times in hours; dividing by 1,000 converts grams to kg. For O the displayed result is converted to kg after summation. For R, N_j are quadrature vehicle weights. Unrounded values are used in the calculations; the substitutions above are rounded.

The speed/rate distinction in this case. The observed mean speed is 23.557 mph and the inferred queue speed is 13.611 mph; both select the same CO_2 lookup rate, 6,931.846 g/(veh h). Free-flow speed selects 20,615.635 g/(veh h). M assigns 23.1% of its vehicle-hours to free flow, which contributes 47.2% of its CO_2 total. Its mean trip consists of 0.570 min at free flow and 1.899 min at queue speed.

Episode emission totals and CO_2 discrepancy

Calculation	CO_2 (kg)	NO_x (g)	CO (g)	HC (g)	CO_2 error (%)
O: Recorded speeds	1,743.79	1,400.82	15,511.48	658.66	Reference
S: Episode mean speed	1,465.63	861.78	13,252.81	581.74	-15.95
R: Reconstructed speeds	1,657.64	1,360.90	14,045.02	609.22	-4.94
M: Two-speed lookup	1,988.14	1,997.71	15,645.69	665.78	+14.01
A: Two-speed cubic	1,838.94	1,712.13	14,960.73	641.44	+5.46

Totals are per lane. Signed error is $100(E_m/E_O - 1)$; its absolute value is the one-episode WAPE. R has the smallest CO_2 discrepancy here. M's absolute discrepancy (14.01%) is smaller than S's (15.95%), unlike the pooled ordering on page 2.

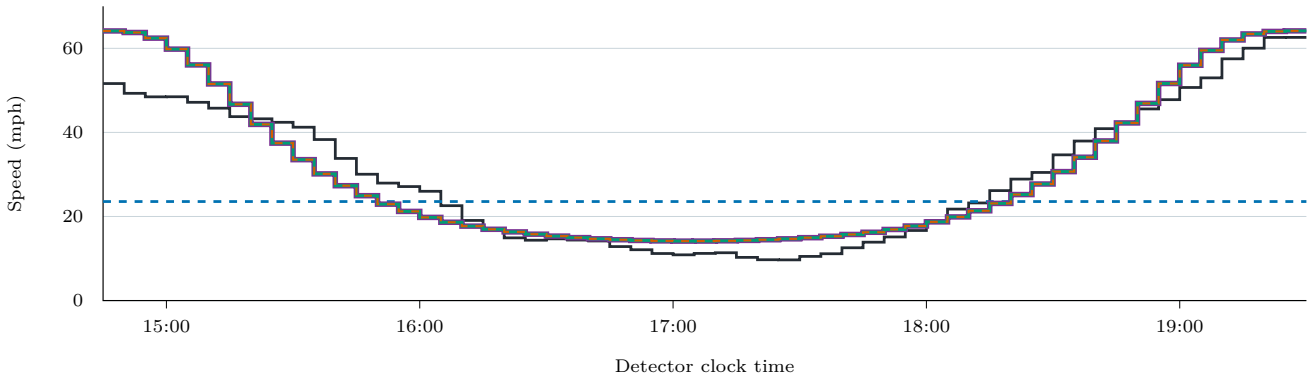
Source: Sensor78_Emission_Calculations.xlsx and its embedded January input profile and calibrated parameters. Recorded source travel times are not used: link times are recalculated as L/v. All emissions use passenger-car rates with zero acceleration and level road.

All five methods on the same clock

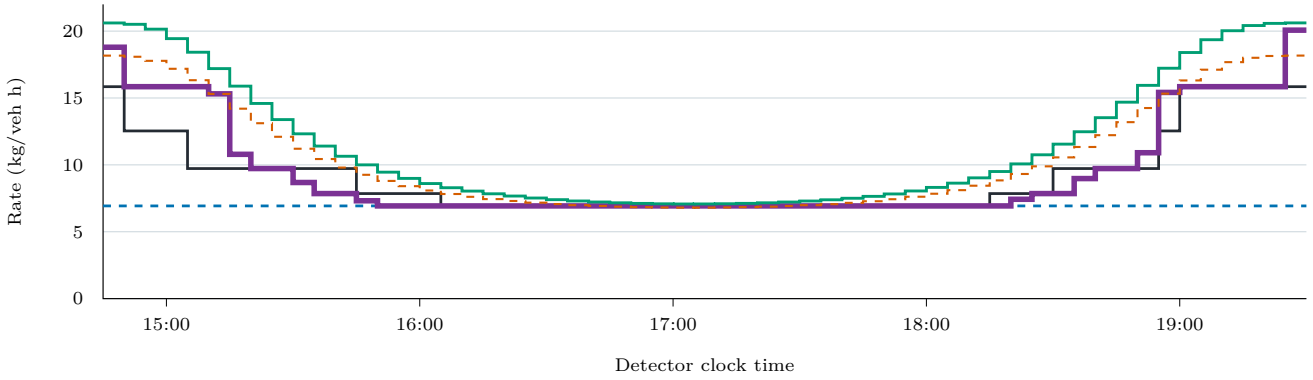
Detector 78. The modeled groups are aligned to the recorded cumulative vehicle counts.

O: recorded S: mean speed R: reconstructed M: two-speed A: cubic

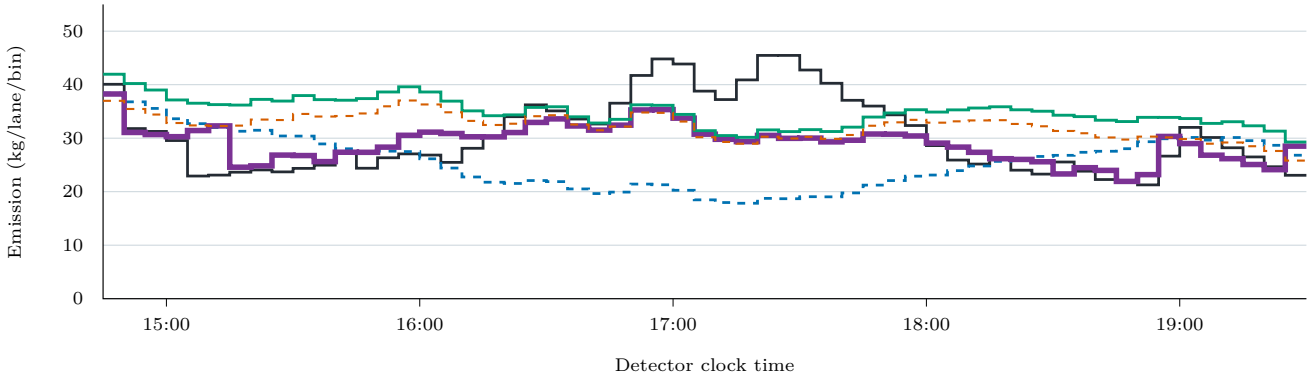
(a) Equivalent whole-link speed: all five approaches



(b) Five-minute effective CO_2 emission rate



(c) CO_2 assigned to the counted vehicles in each five-minute bin

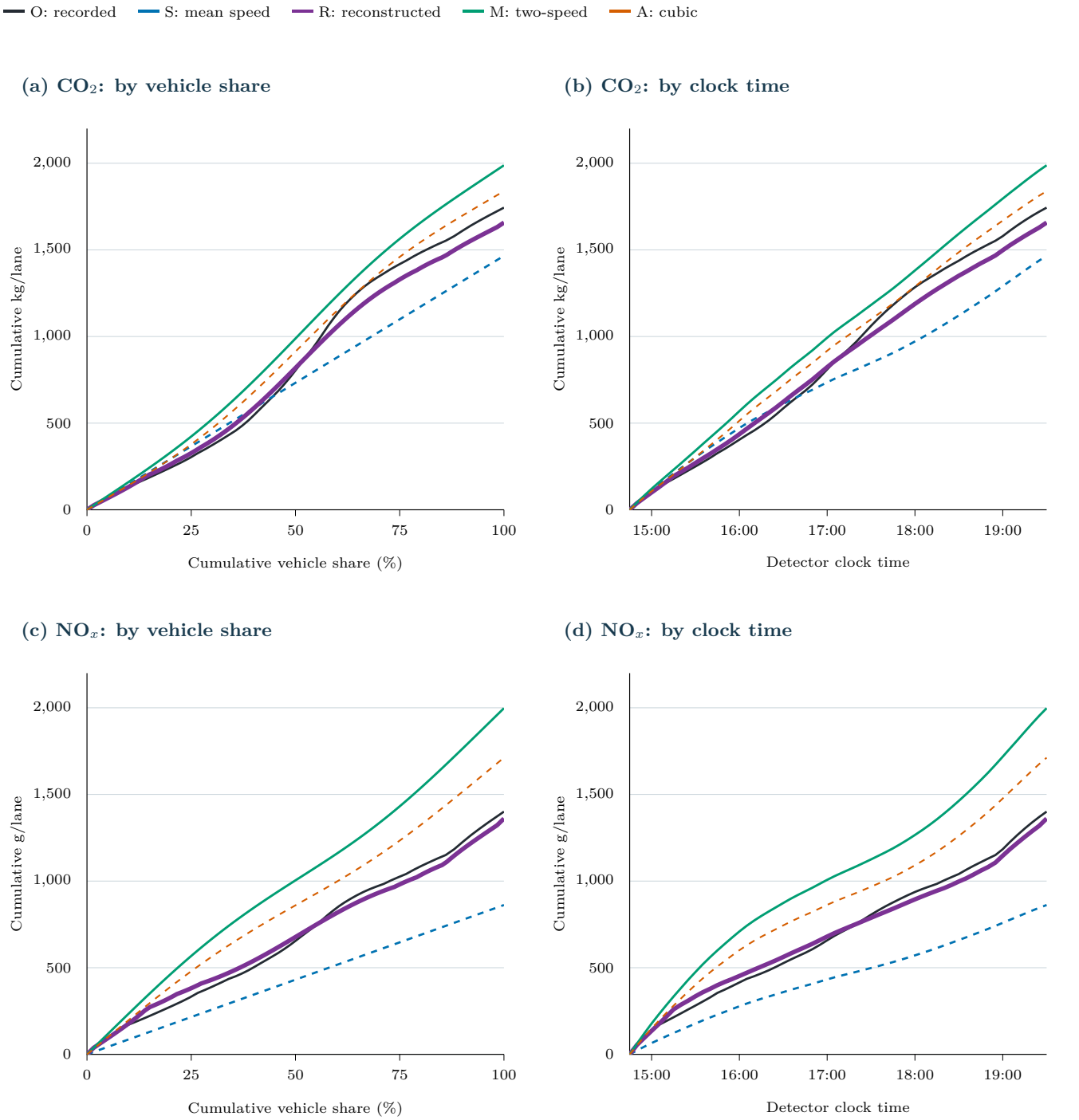


Speed overlap is expected. R, M and A have identical bin vehicle-hours and distance; their equivalent speeds coincide. For M/A, $v_i^{eff} = L/(\bar{T}_{free,i} + \bar{T}_{q,i})$. The colors distinguish emission calculations, not three different equivalent-speed predictions.

For every method, $v_i^{eff} = d_i L/H_i$, $r_i^{eff} = E_i/H_i$ and $E_i = H_i r_i^{eff}$. R's effective rate is the time-weighted rate of its reconstructed groups, not the lookup at its bin-average speed. The displayed emissions are complete-link emissions assigned to vehicles counted in each bin; they are not emissions physically released during that clock interval. The alignment preserves all previous totals.

Cumulative emissions: vehicle share and time

The left panels retain the vehicle-order view; the right panels use the common five-minute clock.



Left: cumulative counted-vehicle share. O follows observed bin order. R, M and A follow modeled group order. S uses a constant emission per vehicle. The same reconstructed group delays supply M/A's free/queue allocation. **Right: detector clock time.** Each boundary is the sum of all preceding five-minute emission contributions. Within-bin lines interpolate the bin contribution for display; no sub-bin release timing is inferred. Every endpoint equals the corresponding total on page 3.

Alignment rule. Split each positive quadrature vehicle weight across observed bins in cumulative-count order. For node j and bin i, the overlap count is

$$n_{ij} = \max\{0, \min(B_i, N_j) - \max(B_{i-1}, N_{j-1})\},$$

where B_i and N_j are cumulative observed-bin and modeled-node counts. Use $H_i = \sum_j n_{ij} T_j$ and $E_i = \sum_j n_{ij} e_j$. Thus all five methods use exactly the same bin count d_i while keeping their original episode totals. This is a transparent display attribution, not an independently reconstructed queue clock.

Where the emission differences enter

An exact signed decomposition for detector 78, conditional on the stated sequence. Define a single-speed calculation using the same fitted mean travel time as R/M/A:

$$E_{S,pred} = H_{pred} r_0(DL/H_{pred}).$$

Here $H_{obs} = 211.434$ and $H_{pred} = 197.040$ veh h/lane, a -6.81% change. Mean trip time changes from 2.649 to 2.469 min and mean speed from 23.557 to 25.278 mph. The latter crosses a lookup-rate boundary; the fitted-mean CO_2 total is 1,547.64 kg/lane. For step 2, an exact symmetric split is

$$\Delta E_H = (H_p - H_o)(r_p + r_o)/2, \qquad \Delta E_r = (r_p - r_o)(H_p + H_o)/2.$$

Here $r_o = r_0(DL/H_o)$ and $r_p = r_0(DL/H_p)$. For CO_2, the vehicle-hours term is -106.42 kg and the mean-speed/rate term is $+188.43$ kg, giving a net $+82.01$ kg. These are mathematical components of the fitted-mean comparison.

Sequential CO_2 contributions

Change	CO_2 difference (kg/lane)	% of O
1. Observed mean-rate aggregation	-278.16	-15.95
2. Fitted mean delay and speed	+82.01	+4.70
3. Reconstructed trip-speed variation	+110.00	+6.31
4. Two operating-speed allocation	+330.50	+18.95
5. Cubic rate approximation	-149.20	-8.56

What each step measures

1. $E_S - E_O$: replace the recorded speed distribution with one observed mean speed, keeping observed vehicle-hours.
2. $E_{S,pred} - E_S$: change to fitted mean delay and its implied mean speed, using the same one-speed rule. This combines the exposure change and the induced lookup-rate change.
3. $E_R - E_{S,pred}$: introduce reconstructed variation among whole-link trip speeds at the same fitted vehicle-hours.
4. $E_M - E_R$: replace each group's whole-trip speed with free/queue operating speeds, preserving time and distance.
5. $E_A - E_M$: replace the two lookup rates with cubic rates; time allocation is unchanged.

Which terms belong to each result? S uses step 1. R uses steps 1–3. M uses steps 1–4. A uses steps 1–5. Their signed sums equal $E_m - E_O$ exactly. For example,

$$E_M - E_O = (E_S - E_O) + (E_{S,pred} - E_S) + (E_R - E_{S,pred}) + (E_M - E_R).$$

Positive and negative components can cancel. Absolute errors or WAPE values do not add this way. This decomposition is order-dependent accounting, not proof of independent causal effects.

Components for all pollutants (percentage points relative to O)

Pollutant	Step 1	Step 2	Step 3	Step 4	Step 5
CO_2	-15.95	+4.70	+6.31	+18.95	-8.56
NO_x	-38.48	+23.90	+11.73	+45.46	-20.39
CO	-14.56	+3.92	+1.18	+10.32	-4.42
HC	-11.68	+3.30	+0.87	+8.59	-3.69

Interpretation. A lower total error can result from cancellation between calibration and emission-representation errors. The current reference itself assumes constant-speed whole-link trips; these components explain discrepancies relative to that reference, not discrepancies from measured tailpipe emissions. The 137-episode analysis will assess these patterns across episodes separately.